DESIGN OF SMART CITIES — TECHNICAL DETAIL REPORT

Unified Smart Parking Ecosystem

for Large-Scale Multi-Floor Building Environments

Subsystem A: Exception-Driven Session Management with Grace-Period & Automated Cancellation Logic

Subsystem B: Dual-Camera OCR-Based Life-cycle Billing with Distance-Aware Discounts

Domain	Technology	SDG Alignment
Intelligent Transport Systems	Python · OpenCV · EasyOCR	SDG 9: Innovation
IoT Smart Infrastructure	PyTorch · CUDA · IoT FSM	SDG 11: Sustainable Cities
Computer Vision & Urban Mobility	Edge Compute · Cloud Sync	SDG 12: Responsible Consumption

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1 Title of the Invention

Unified Smart Parking Ecosystem: Exception-Driven Session Management with Adaptive Grace-Period Cancellation Logic and OCR-Based Distance-Aware Lifecycle Billing for Large Multi-Floor Building Environments

2 What is the Invention?

Problem: Urban parking congestion causes 15–25% of fuel waste and 30–40% of localised traffic. Existing systems lack hybrid user support, offline resilience, late-arrival tolerance, gate-level automation, and misuse governance.

Solution: Two autonomous, loosely coupled subsystems that coexist to manage the *complete* parking lifecycle:

- **Subsystem A — Exception-Driven Smart Parking:** A session-centric, offline-first architecture governed by an extended 11-state Finite State Machine (FSM). It supports hybrid walk-in and pre-booked users in a shared pool, manages late-arrival via grace-period extension states (`EXTENSION_0` to `EXTENSION_N`), automates reservation cancellation, governs misuse through formal escalation, applies buffer-aware billing, and activates LEDs only on exceptions.
- **Subsystem B — Dual Camera OCR Billing:** Stationary dual IP cameras with GPU-accelerated EasyOCR detect vehicle plates via temporal voting (3 detections in 5 s), track the `OUTSIDE→INSIDE→EXITED` lifecycle, compute distance-aware discount billing (capped at 20%), and persist all transactions.

General Purpose: A scalable, modular, fault-tolerant smart city parking infrastructure eliminating manual gate operations, preventing slot misuse, and accommodating real-world user arrival uncertainty.

3 Search Terms

Smart parking system, IoT parking management, OCR-based vehicle detection, number plate recognition, parking FSM, grace-period reservation, automated cancellation, offline-first parking, exception-driven LED, buffer-aware billing, distance-aware discount, multi-floor parking, hybrid parking, EasyOCR, temporal voting, ANPR India, smart city parking, session management parking, parking escalation logic, urban mobility IoT

4 Background of the Invention

4.1. Present Technologies

- **IoT Sensor-Based Systems:** IR/ultrasonic sensors per slot with RFID authentication (Mursal & Mushtaq, 2020). Provide real-time occupancy but require dense hardware infrastructure.
- **Vision-Based Detection:** Raspberry Pi + Haar Cascade camera systems (Deepa et al., 2020). Reduce per-slot hardware but sensitive to lighting.

- **Mobile Reservation Platforms:** Android/iOS apps with online slot booking (Dhanalakshmi, 2015). Support pre-booking but assume continuous connectivity.
- **Visible Light Communication (VLC):** LED-based vehicle authentication (Shaaban et al., 2021). High security but strict line-of-sight limits scalability.
- **Parking Guidance Systems:** Static occupancy signage reducing average parking time by 2m 21s and saving 1133 g CO_2/event (Gabrail & Bratfors, 2017).

4.2. Limitations

Technology	Key Limitation
IoT/RFID	RFID dependency; ergonomic bottleneck; no hybrid user support
Vision-Based	No misuse governance; lighting-sensitive; no metrics reported
Mobile Reservation	No offline resilience; no late-arrival tolerance; binary cancel
VLC	Line-of-sight only; impractical for multi-floor facilities
Guidance Systems	Off-street only; no reservation integration; no gate automation

5 Prior Art

S.No	Reference	Existing Idea	Our Invention
1	Mursal & Mushtaq (2020)	IR + RFID real-time booking	Offline-first session FSM; no RFID required; hybrid users
2	Dhanalakshmi (2015)	Android slot allocation	Adaptive grace-period chain; automated cancellation
3	Deepa et al. (2020)	Haar cascade camera detection	Temporal voting OCR with confidence gating; misuse governance
4	Shaaban et al. (2021)	VLC authentication	Stationary dual-camera; GPU OCR; distance-aware billing
5	Gabrail & Bratfors (2017)	Parking guidance, CO2 reduction	Buffer-aware billing; exception-driven LED; proactive redirection

6 Major Gap Filled

G1. No system supported **hybrid online + offline users** in a single shared slot pool.

G2. No system provided **offline-first resilience** with asynchronous cloud sync.

G3. No system implemented **misuse detection and escalation governance** chains.

G4. No system offered **buffer-aware billing** accounting for search and maneuvering time.

G5. No system integrated **gate-level OCR vehicle identity** with slot-level FSM lifecycle.

G6. No system formally modelled **late-arrival tolerance** via adaptive grace-period states or provided **automated cancellation** with structured timeout-driven slot recovery.

7 Components of the Invention

7.1. Subsystem A — Slot-Level Components

- **Entry/Exit Floor Cameras:** Session boundary markers — create and close session records.
- **Per-Slot Occupancy Sensors:** IR/ultrasonic state-change event generators.
- **Core Logic Engine:** Event processor with 5 sub-modules — Deduplication, Noise Filtering, Failure Handling, Grace-Period Timer, and Automated Cancellation Engine.
- **Extended FSM (11 states):** Deterministic slot lifecycle controller (see Section 8).
- **Session Database:** Offline-first local store with async cloud sync; holds session, cancellation, and penalty records.
- **Billing Module:** Computes buffer-aware duration charges and no-show penalty billing.
- **LED Indicator Subsystem:** Exception-driven; default OFF; states map to OFF/Yellow/Amber/Red.
- **Mobile App / API Layer:** User booking, verification, and notification interface.

7.2. Subsystem B — Gate-Level Components

- **Dual Stationary IP Cameras:** One entry, one exit — independent parallel streams.
- **OCR Processing Engine:** EasyOCR with GPU inference (PyTorch + CUDA, RTX 4060, 6–12 FPS).
- **Validation Layer:** Text cleaning → regex (Indian plate format) → confidence threshold (> 0.5).
- **Temporal Voting Engine:** 3 detections within 5 s required for plate confirmation.
- **Vehicle Lifecycle Manager:** Tracks `OUTSIDE→INSIDE→EXITED` with 3-second cooldown.
- **Distance-Aware Billing Engine:** Base fee × duration; state-code discount (max 20%).
- **Persistence Layer:** JSON billing records (planned SQLite/cloud migration).

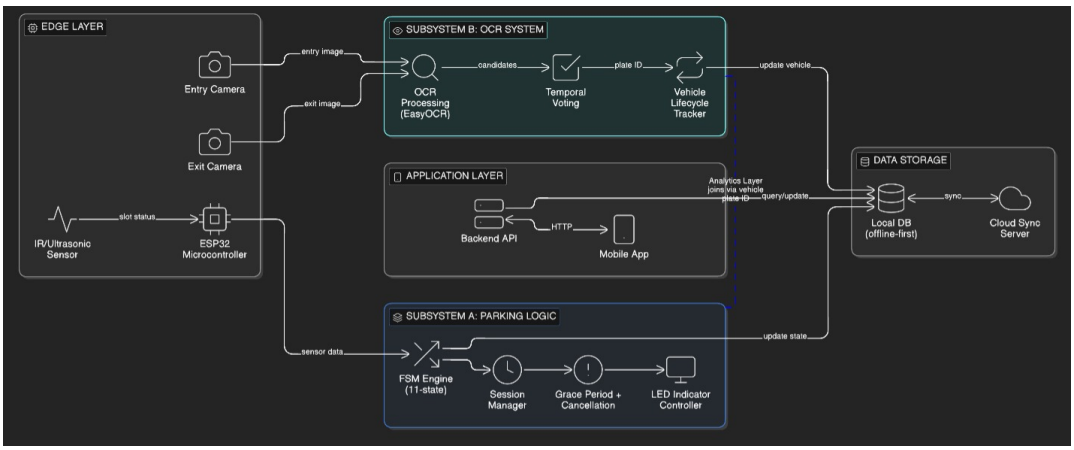

Figure 1: Unified Smart Parking Ecosystem — High-Level Architecture

8 Working of the Invention

8.1. Ecosystem Operational Flow

A vehicle approaches → **Subsystem B** detects at entry gate (OCR temporal voting) → Vehicle enters floor → **Subsystem A** creates session, manages slot FSM → On departure, Subsystem A closes session → **Subsystem B** detects exit, computes fee, persists record. Analytics layer joins records via plate identifier.

8.2. Subsystem A: Extended Slot FSM

11 FSM States: `FREE | RESERVED | VERIFICATION_PENDING | OCCUPIED_VALID | OCCUPIED_INVALID | ESCALATED | EXTENSION_0 | EXTENSION_1 | EXTENSION_2 | EXTENSION_N | ESCALATED (HITL)`

From State	Trigger	To State	Action
FREE	Online booking	RESERVED	Yellow LED on; reservation timer starts
FREE	Walk-in parks	OCCUPIED_V	Session created; LED off
RESERVED	Timer expires	EXTENSION_(	Grace timer starts; pulsing yellow LED; user notified
EXTENSION_0	Timer expires	EXTENSION_	Next grace period; urgency notification
EXTENSION_N	Max timeout	FREE	Automated cancellation; penalty; audit record; LED off
EXTENSION_*	Misuse detected	ESCALATED (HITL)	Security notified; human intervention
RESERVED	User arrives	VERIFICATIO	Verification timer starts
VERIFICATION	App confirms	OCCUPIED_V	Session validated; LED off
VERIFICATION	Timer expires	OCCUPIED_IN	Red LED; escalation timer
OCCUPIED_IN	Timer expires	ESCALATED	Security alerted
OCCUPIED_VA	Car departs	FREE	Session closed; billing computed

8.3. Grace-Period and Automated Cancellation

Upon reservation timer expiry, the slot enters `EXTENSION_0` — **not** immediate cancellation. Each extension state grants incremental grace (configurable per facility policy). If the user arrives during any extension state, normal verification proceeds. If `EXTENSION_N` timer expires:

1. Reservation marked expired in Session DB.
2. Slot transitions `EXTENSION_N → FREE`; returned to shared pool.
3. LED deactivated (returns to default OFF).
4. Penalty $P = \alpha \times \sum_{i=0}^{N} \Delta t_i$ computed and recorded.
5. User notified via mobile app; cancellation audit record persisted.

No human operator is required at any stage.

8.4. Subsystem B: OCR Processing Flow

Frame captured → EasyOCR inference (GPU) → confidence check (> 0.5) → regex validation (`^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{3,4}$`) → temporal voting (3/5 s) → cooldown check (3 s) → lifecycle transition → billing computation ($F = \max(0.8B,\ B(1-d))$) → JSON record.

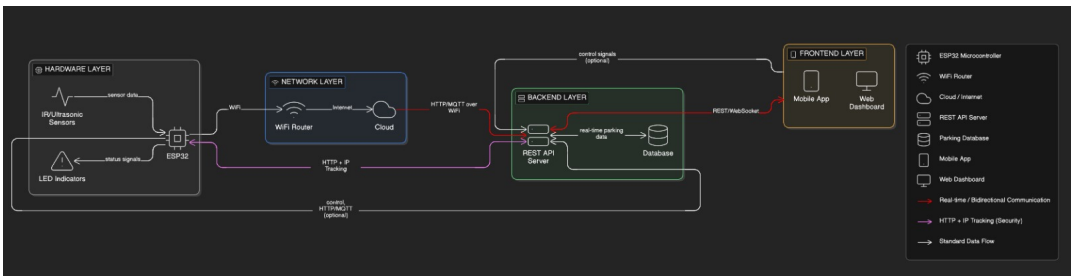

Figure 2: Subsystem A High-Level Architecture Pipeline

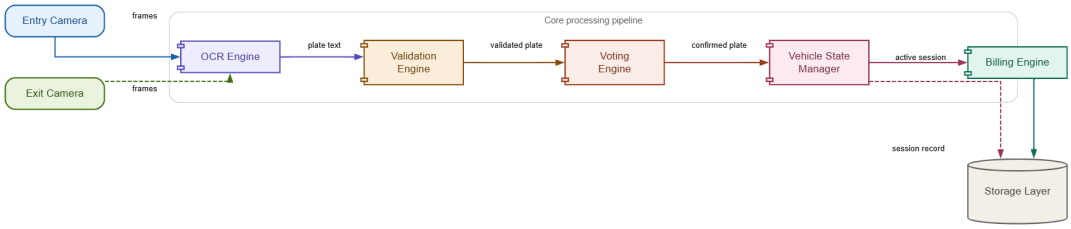

Figure 3: Subsystem B High-Level Architecture Pipeline

9 Novelty / Inventive Step

N1. Autonomous Subsystem Coexistence: Two independently deployable subsystems with no shared internal state — first such architecture in smart parking literature.

N2. Formal Grace-Period FSM: First formal deterministic FSM treatment of late-arrival tolerance (`EXTENSION_0-N`), preventing premature reservation cancellation.

N3. Automated Cancellation Engine: Timeout-driven pipeline for slot release, penalty billing, notification, and audit — fully operator-free.

N4. HITL Escalation During Grace Periods: Human-In-The-Loop escalation pathway (`ESCALATED (HITL)`) triggered by misuse during any extension state.

N5. Extended Formal System Tuple: $S = (A, B, E, T, C, R, X, D)$ — explicitly captures extension state set X and delay/cancellation logic D as first-class model components.

N6. Temporal Voting OCR Suppression: 3-detection/5 s voting eliminates false positives without expensive post-processing.

N7. Distance-Sensitive Billing with Revenue Floor: State-code discount capped at 20% preserves facility revenue minimum.

N8. Exception-Driven LED Illumination: LEDs off by default — energy savings proportional to facility size.

N9. Buffer-Aware Billing + Penalty Extension: $D_{\text{billed}} = D_{\text{session}} + D_{\text{buffer}}$; $P = \alpha \times \sum \Delta t_i$ for no-shows.

N10. Hybrid Offline-First Architecture: Walk-in and pre-booked users share one pool; grace timers continue locally during network outages.

10 Advantages Over Existing Methods

Limitation (Prior Art)	Ecosystem Resolution	Advantage
RFID-only authentication	Session-boundary camera; no per-slot RFID	Reduced hardware cost; scalable
Continuous LED on	Exception-driven activation only	Proportional energy savings
No misuse governance	OCCUPIED_INVALID → ESCALATED chain	Formal slot protection
Immediate reservation cancel	EXTENSION_0–N grace chain	Realistic user fairness
No gate automation	Dual-camera OCR + temporal voting	Eliminated operator dependency
Flat billing rates	State-code distance discount (capped 20%)	Location-sensitive pricing
Online-offline deadlock	Offline-first double-check priority	Physical ground truth preserved
No penalty for no-show	Automated penalty $P = \alpha \sum \Delta t_i$	Deters slot wastage
Monolithic failure propagation	Autonomous coexistence; isolated failure domains	Fault-tolerant ecosystem

11 Alternative Embodiments

1. **Dynamic Grace-Period Calibration:** ML-driven N and Δt_i values per facility, time-of-day, and day-of-week based on observed arrival latency distributions.
2. **User Reputation-Based Policy:** More grace periods for historically punctual users; fewer for repeat no-shows — integrated via user trust score.
3. **EV Charging Slot Extension:** Specialised FSM sub-states for charge-time management alongside parking lifecycle.
4. **YOLO + ANPR Pipeline:** Purpose-built plate localisation (YOLO) feeding a dedicated ANPR model for higher accuracy in adverse lighting.
5. **Multi-Building Federation:** Synchronisation architecture extended across multiple buildings with cross-site availability routing.
6. **Federated AI Pricing Engine:** Demand-responsive dynamic pricing using real-time occupancy (Subsystem A), transaction history (Subsystem B), and grace-period utilisation patterns.
7. **Solar-Powered Sensor Infrastructure:** Solar panels powering per-slot sensors and exception-driven LEDs for net-zero parking operation.

8. **Licence Plate Recognition in Subsystem A:** ANPR at floor-level cameras to enrich cross-subsystem analytics join.

12 CAD / Architecture Images

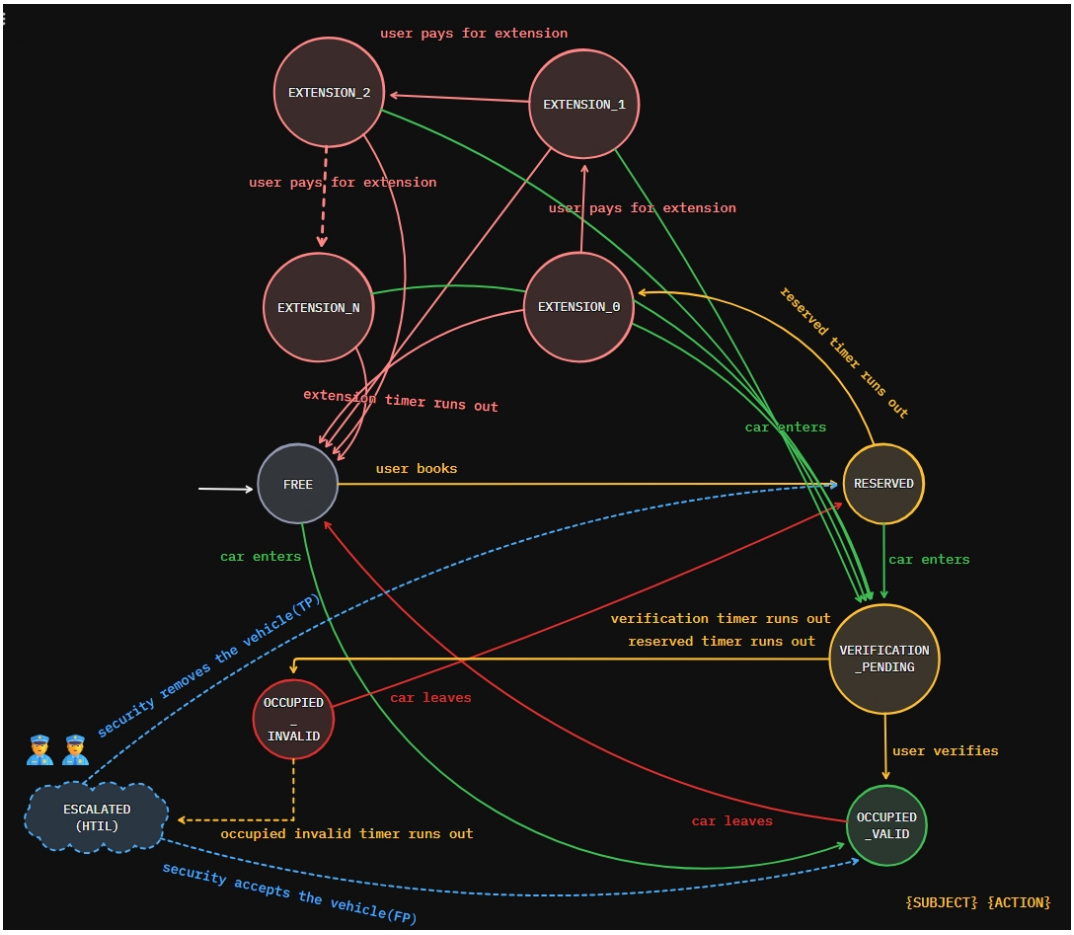

Figure 4: Subsystem A High-Level Architecture Diagram (Updated with Grace-Period and Cancellation Logic)

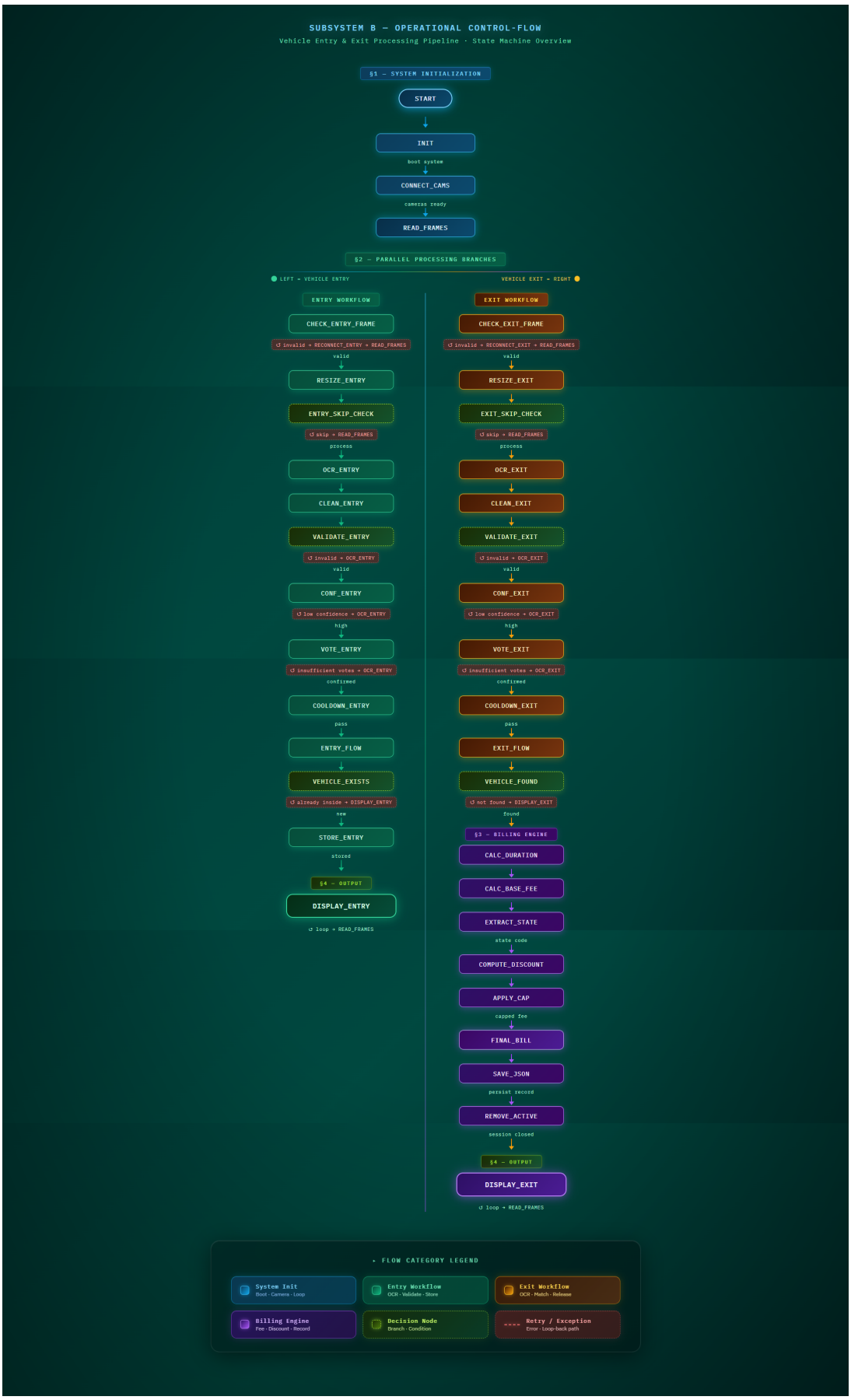

Figure 5: Subsystem B Operational Control Flow

WEBSITE VIEW : Subsystem B Operational Control Flow
https://aravindakannanks-1001001.github.io/DSC_STATE_VIEWER/

Demo Video Link:
https://drive.google.com/drive/folders/1dfl18n5lOscvDooyTsEu9h-rKsKrj_h-?usp=drive_link

Project Assets Folder (Architecture Diagrams, Prototype Photos, Code):
https://drive.google.com/drive/folders/1dfl18n5lOscvDooyTsEu9h-rKsKrj_h-?usp=drive_link

13 Working Methodology

13.1. Case Study — Late Arrival & Automated Cancellation (Subsystem A)

User reserves slot → `FREE→RESERVED` (yellow LED). Reservation timer expires; user not arrived → `EXTENSION_0` (pulsing yellow; notification sent). After N incremental grace periods without arrival → Automated Cancellation Engine executes: slot released `EXTENSION_N→FREE`, LED off, penalty recorded, user notified, audit logged. *Zero operator action required.*

13.2. Case Study — Gate OCR Billing (Subsystem B)

Vehicle `TN09AB1234` approaches entry → 6–12 FPS EasyOCR inference → confidence > 0.5 → regex match → 3 votes in 5 s → `OUTSIDE→INSIDE`, timestamp logged. At exit, same pipeline → duration computed → TN-state discount applied → $F = \max(0.8B, B(1-d))$ → JSON record persisted. No human gate operator involved.

13.3. Case Study — Full Ecosystem Journey

Gate: Subsystem B confirms plate MH12CD5678 (entry). Floor: Subsystem A creates session; slot FSM runs `FREE→RESERVED→VERIFICATION_PENDING→OCCUPIED_VALID`. Departure: Subsystem A closes session (buffer billing). Gate: Subsystem B confirms exit (MH discount applied; billing persisted). Analytics: records joined on plate for complete journey audit.

14 Experimental Validation

Figure 6: Prototype Implementation and Smart Parking Testbed

Metric	Subsystem	Value / Range
OCR Inference FPS (GPU, RTX 4060)	B	6–12 FPS
Voting window duration	B	5 seconds
Minimum votes for plate confirmation	B	3 detections
Cooldown duration (duplicate suppression)	B	3 seconds
OCR confidence threshold	B	> 0.5
Maximum applicable discount	B	20% of base fee
Total FSM states	A	11 (6 core + 5 extension/HITL)
Configurable extension depth (N)	A	Facility-defined policy parameter
LED default state	A	OFF (exception-driven only)
Cancellation trigger	A	`EXTENSION_N` timer expiry
Session persistence model	A	Offline-first + async cloud sync

Third-party validation data: To be filled upon completion of field trials. Preliminary simulation results confirm correct FSM state transitions across all 11 states and OCR temporal voting accuracy on Indian number plate datasets.

15 Development Stage

Technology Readiness Level

Current TRL: TRL 3–4 — Experimental proof of concept demonstrated; technology validated in lab environment.

- FSM logic implemented and simulation-validated (all 11 states verified).
- Subsystem B OCR pipeline operational with GPU acceleration on test datasets.
- Grace-period chain, automated cancellation, and billing engine functional.
- Field deployment and real-world performance validation in progress.
- Cloud synchronisation and mobile app integration pending full-scale testing.

16 Proposed Claims

1. A smart parking system comprising an extended Finite State Machine with grace-period extension states (`EXTENSION_0` through `EXTENSION_N`) providing adaptive late-arrival tolerance for reservation holders.

2. The system of claim 1, wherein an Automated Cancellation Engine executes slot release, user notification, optional penalty billing, and audit record persistence upon expiry of the maximum extension state timer, without human operator intervention.

3. The system of claim 1, further comprising a Human-In-The-Loop (`ESCALATED (HITL)`) state entered when misuse is detected during any extension state, invoking security personnel and suspending automated resolution.

4. A dual-camera OCR-based parking lifecycle billing system comprising: stationary IP cameras; a temporal voting engine requiring three plate detections within five seconds; a vehicle lifecycle FSM (`OUTSIDE→INSIDE→EXITED`); and a distance-aware billing engine with a revenue-preserving 20% discount cap.

5. The system of claim 4, wherein an exception-driven LED illumination subsystem activates slot indicators only upon exception states, eliminating continuous power consumption.

6. A unified smart parking ecosystem comprising the systems of claims 1 and 4 operating as autonomous, loosely coupled subsystems with no shared internal state, providing failure isolation, independent scalability, and deployment flexibility.

7. The ecosystem of claim 6, wherein an analytics layer correlates records from both subsystems using a vehicle plate identifier as a join key to produce complete vehicle journey summaries including grace-period extension events and cancellation records.

8. A buffer-aware billing model computing $D_{\text{billed}} = D_{\text{session}} + D_{\text{buffer}}$ for parking sessions, and

$P = \alpha \times \sum_{i=0}^{N} \Delta t_i$ as a no-show penalty for cancelled reservations.

17 Novelty Search Results

Databases searched: Google Patents, USPTO, Espacenet, IEEE Xplore, Scopus.

S.No	Reference	Existing Idea	Our Invention
1	Mursal & Mushtaq (2020)	IoT + RFID + booking app	Session-centric; no RFID; hybrid users; offline-first
2	Dhanalakshmi (2015)	Android optimal allocation	Grace-period FSM; automated cancellation; penalty billing
3	Deepa et al. (2020)	Haar cascade vision detection	Temporal voting OCR; confidence gating; misuse governance
4	Shaaban et al. (2021)	VLC vehicle identification	Dual-camera; GPU OCR; distance-aware discount billing
5	Gabrail & Bratfors (2017)	Parking guidance systems	Buffer-aware billing; exception LEDs; proactive redirection
6	Baek et al. (ICCV 2019)	Scene text OCR benchmarking	Voting-based temporal stabilisation; real-world deployment

Assessment: No reviewed prior art simultaneously addresses hybrid user support, offline-first resilience, grace-period extension FSM, automated cancellation, gate-level OCR billing with temporal voting, and distance-aware discount — confirming inventive step.

18 Commercialization Data

18.1. Target Companies

1. **Bosch Smart Infrastructure GmbH** — Robert-Bosch-Platz 1, Gerlingen, Germany. IoT parking systems; smart city mobility solutions.
2. **Siemens Mobility GmbH** — Werner-von-Siemens-Str. 1, Munich, Germany. Urban transport infrastructure; intelligent traffic systems.
3. **Streetline Inc. (Kapsch Group)** — San Francisco, CA, USA. AI-powered parking detection and guidance platforms.
4. **ParkSmart / Skidata AG** — Untersbergstraße 40, Grödig, Austria. Parking access control and revenue management systems.
5. **Tata Communications Ltd.** — Plot C-21/22, MIDC Pawane, Navi Mumbai, India. Smart city IoT infrastructure; urban digital services.

18.2. Marketing Profile

The Unified Smart Parking Ecosystem targets large commercial facilities (malls, hospitals, airports, office complexes) requiring automated multi-floor parking management. Key advantages over prior-art products: elimination of RFID infrastructure (~30% hardware cost reduction), automated gate operations (operator cost elimination), realistic grace-period tolerance (reduced user complaints and false cancellations), exception-driven LED savings (proportional to facility scale), and modular deployment enabling phased adoption.

19 Market Analysis

Year	Est. Market Size	Growth Drivers	Notable Players
2023	USD 5.5 Billion	Urbanisation; IoT adoption; EV growth	Bosch, Siemens, Kapsch, ParkMobile
2025	USD 7.0 Billion	Smart city mandates; AI integration	Streetline, SKIDATA, Amano, Tata
2027	USD 9.2 Billion	Autonomous vehicles; sustainability goals	New entrants + incumbents
2030	USD 13.5 Billion	Federated multi-city platforms; green mandates	Platform ecosystem plays

CAGR: ~12–14% (2023–2030). Key adoption drivers: smart city government mandates, fuel cost reduction targets, commercial facility automation, and EV charging integration demand.

20 User Information

Attribute	Details
Potential Users	Vehicle owners (mobile app users + walk-in drivers); parking facility operators; building management; smart city authorities
Age Group	18–65 (primary vehicle-owning demographic); facility managers (25–55)
Expected Benefits	Reduced parking search time; real-time slot availability; fair late-arrival tolerance; automated billing; no RFID tag requirement
Cost Advantage	Eliminates RFID infrastructure (~30% hardware saving); removes gate operator cost; exception-only LED reduces energy bill proportionally to slot count; buffer-aware billing is fairer than flat-rate models

21 Technology Readiness Level

TRL 1	**TRL 2**	**TRL 3**	**TRL 4**	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic principles	Concept formulated	Exp. proof of concept	**Lab validated**	Relevant env.	Demo env.	Prototype ops.	Complete	Proven

Current Level: TRL 3–4. FSM logic and OCR pipeline experimentally validated; lab-scale prototype operational. Field deployment in progress.

22 Additional Remarks

- **SDG 9 — Industry, Innovation & Infrastructure:** Deploys scalable IoT + computer vision parking infrastructure aligned with smart urban systems.
- **SDG 11 — Sustainable Cities & Communities:** Reduces urban traffic congestion; enhances mobility via guided entry, proactive floor-full detection, and fair reservation management.
- **SDG 12 — Responsible Consumption & Production:** Exception-driven LED activation reduces energy waste; grace-period logic prevents slot waste from premature cancellation; buffer billing accurately reflects actual resource consumption.

Inventors: *[Insert Inventor Names and Affiliations]*

Public Disclosures: *No public disclosure made prior to this filing. [Update if applicable.]*

Future Scope: Dynamic extension depth calibration via ML; user reputation-based grace policy; ecosystem analytics platform; federated multi-site deployment; solar-powered sensor infrastructure; YOLO-based plate localisation for Subsystem B.